Final Report 499A– Group-1

Project: ***Time-Series Few-Shot Anomaly Detection***

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Report: Few-Shot Anomaly Detection Methods and Their Evaluation

1. Method

FewSOME,LSTM-AE for HVAC Systems,TAMA (Time series Anomaly Multimodal Analyzer),FastRecon,MOSAD (Multivariate Open-Set Anomaly Detector),Adversarial FSAD,DACAD (Domain Adaptation Contrastive Learning for Anomaly Detection).

2. Dataset

1.FewSOME: MNIST, CIFAR-10, Fashion-MNIST, MVTec AD

2.LSTM-AE: MZVAV-2 dataset from ASHRAE 1312

3.TAMA: Web services (SMD), industry (UCR), healthcare (ECG), and transportation (Dodgers).

4.FastRecon: MVTec AD, MPDD

5.MOSAD: Server Machine Dataset (SMD), Physikalisch-Technische Bundesanstalt XL (PTB-XL), Temple University Hospital Electroencephalogram Seizure Corpus (TUSZ)

6.Adversarial FSAD: MVTec AD, DAGM2007
7.DACAD: MSL (Mars Science Laboratory), SMAP (Soil Moisture Active Passive), SMD, Boiler Fault Detection.

3. Evaluation

AUC (Area Under the Curve),F1 Score,Precision-Recall Curve,True Positive Rate (TPR) / False Positive Rate (FPR),Balanced Accuracy,
Pixel-Level AUROC (Area Under Receiver Operating Characteristic Curve).

4. Technology

Siamese Networks,LSTM Autoencoders,Multimodal Models (e.g., GPT-4o, TAMA),ResNet-50,Temporal Convolutional Networks (TCN),Contrastive Learning.

5. What Was Learned

- Few-Shot Learning Advantage: The methods introduced in these papers demonstrate the power of few-shot learning in anomaly detection, significantly reducing the amount of labeled data needed, which is crucial in real-world applications where labeled anomalies are scarce.

- Stop Loss in FewSOME: The introduction of the "Stop Loss" technique in FewSOME improves model robustness and prevents representational collapse when training on a small number of samples.

- Domain Adaptation in HVAC: The LSTM-AE-based approach for HVAC anomaly detection shows that few-shot learning can be used effectively for domain adaptation, making it suitable for real-world industrial systems with limited labeled data.

- Multimodal and Contrastive Learning: The use of multimodal models (such as TAMA) and contrastive learning (used in MOSAD and DACAD) helps to improve anomaly detection performance, even when dealing with complex, multivariate, or noisy datasets.

Conclusion

The studies reviewed in the paper provide innovative approaches to few-shot anomaly detection, showcasing the effectiveness of deep learning techniques like Siamese networks, LSTM autoencoders, and contrastive learning. These methods are shown to perform well across a variety of datasets and domains, making them highly applicable for industries dealing with limited labeled data. The integration of adversarial loss and domain adaptation further enhances the robustness and generalization of anomaly detection models.